Splash Screen

PROJECT
PROPOSAL

The Ripple Effect

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Problem domain

South Africa faces an ongoing water scarcity crisis, with an average annual rainfall of only 464 mm, significantly below the global average of 860 mm. This challenge is particularly present in urban and suburban areas, where the municipal restrictions and higher water tariffs are common. There are an estimated 800,000 residential swimming pools in South Africa, each requiring between 30, 000 to 60, 000 litres of water to function. The problem focus is not the existence of pools, but the inefficient, uncoordinated, and technically inconsistent approach to their maintenance.

Maintenance complexity

Many new pool owners often find balancing the pH, chlorine and alkalinity challenging. While seasoned veterans are more knowledgeable, natural human errors such as forgetting to add chlorine or reset the water pump due to load shedding can lead to issues including algae growth, damaged equipment, and the waste of a critical resource: water. In addition to these general issues, local challenges also exist. Municipal water regulations are in place in many areas of South Africa, and pool owners face limits on topping up water levels.

Time Constraints

Pool maintenance, if performed correctly, is not the most time-consuming household activity. However, when paired with a business work schedule, other household chores and unforeseen daily occurrences, many pools are neglected. Frequent travellers, holiday accommodation owners and busy residents may not have the time to carry out regular maintenance tasks themselves. As a result, they often rely on professional pool service providers.

Scattered Service Market

As referenced in the previous segment of this analysis, pool maintenance can be approached in two manners: self-service and professional-service outsourcing. Both models require pool service products and for homeowners, an optional need for professional services that are verified and reliable. For local businesses, while there are platforms such as company websites, Facebook Marketplace, and other similar applications, these are often decentralized from the client and the business at first unless a client or business starts the interaction. For the homeowner, finding a reliable and vetted local product or service can be difficult.

The problem isn’t that there are no solutions but rather it’s that they’re scattered, and none are tailored to South Africa’s unique challenges. A solution that combines DIY water quality monitoring, digital maintenance guidance, and a transparent, centralised service marketplace could simultaneously reduce water waste, lower costs, and improve service reliability.

Proposed solution

Our proposed system, *SplashScreen*, is a mobile application developed in Java using Android Studio, with Firebase Firestore as the backend database. It is designed to tackle the core challenges faced by swimming pool owners: maintenance complexity, time constraints, and a scattered service marketplace. By providing a single, data-driven platform, *SplashScreen* will help users perform accurate and efficient maintenance while connecting them with trusted local service providers. This unified approach will replace fragmented tools and word-of-mouth searches, ultimately reducing water waste, lowering maintenance costs, and improving service quality.

For pool owners, the app will offer:

- Registration and tracking of one or more pools.
- Personalised maintenance guides with weekly and monthly checklists.
- A water chemistry calculator that gives precise dosing instructions based on pool size and manually added readings.
- Direct booking of vetted service providers and in-app ordering of pool chemicals and equipment.

For service providers, it will include:

- Business profiles with services, pricing, and availability.
- A searchable marketplace for customers to book and review services.
- Job management tools, real-time chat with clients, and order tracking.

The system will be constructed based on a **14-table Firestore database**, including collections for users, roles, pools, maintenance logs, water readings, providers, bookings, products, weather data, restrictions, and notifications. A secure role-based login will ensure users only see functions relevant to them. The app will integrate external APIs for live weather and load shedding data, generating automated, context-aware recommendations.

Key Screens (Functionality to be extended)

- <u>Login & Registration Screen</u> for secure authentication.
- <u>Dashboard</u> tailored to the user's role, summarizing key data and actions.
- <u>Service Booking Screen</u> to request, view, and manage appointments.
- <u>Maintenance Log Screen</u> for recording and viewing pool water quality data.
- <u>Inventory Management Screen</u> for tracking products and equipment.
- <u>Reports Screen</u> to view or export service and maintenance summaries.
- <u>Help & Support Tips</u> offering guidance, FAQs, and contact details.

As **The Ripple Effect**, *SplashScreen* brings together easy-to-use self-maintenance tools and a trusted services marketplace, designed specifically for the needs of South African pool owners. By tackling challenges like water scarcity, busy schedules, and unreliable service access, it offers a secure platform backed by GitHub version control and organised through ClickUp project management.